WARAITIP LAOSANGPRATEEP

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Thailand / South Korea / Singapore / APAC · Available now

Dual-degree engineer with a Chemical Engineering foundation (SIIT, Thailand) and a Master's in Medical Engineering (University of Auckland), backed by research experience in New Zealand, Thailand, and Taiwan. Combines process engineering fundamentals, hands-on materials characterisation, and applied data science, including machine learning and statistical analysis using Python, SQL, and Power BI. Hands-on experience collaborating across multicultural teams and adapting quickly to different working environments, alongside an interest in medical devices, materials science, and quality engineering, and a growing focus on health, beauty, and wellness, particularly across South Korea and the broader APAC region.

EDUCATION

Master of Medical Engineering Referee — Mar 2025 – Jun 2026

University of Auckland · New Zealand · GPA 7.00 / 9.00 (Graduated with Distinction)

- Thesis: Quantitative mechanical characterisation of neonatal lamb growth plate tissue via nanoindentation/microindentation, mapping stiffness and hardness across four structurally distinct zones (articular, hypertrophic, proliferative, resting)
- Applied statistical testing to isolate significant results from high-variance datasets, reconciling findings against published literature
- Managed the full experimental pipeline — cryogenic tissue prep, controlled conditions, data acquisition, and outlier analysis — to produce publication-quality figures

Coursework Project — Machine Learning for Materials Characterisation (Python & Supervised Learning)

- Built and validated SVM classification models (Python, cross-validation) to classify meat types from spectral wavelength data
- Applied data preprocessing and model evaluation for materials and process data interpretation

Coursework Project — Electric Vehicle Charging Infrastructure Business Case (Auckland Region)

- Developed a Single Stage Business Case (SSBC) for a regional EV charging programme, assessing strategic, financial, and operational feasibility
- Ran multi-criteria analysis against investment objectives: EV adoption, grid efficiency, transport integration, and equitable access
- Assessed delivery models, funding pathways, and risk for investment

- Coursework: Advanced Functional Materials · Battery & Superconductor · Semiconductor & Materials Science · Medical Device & Technology Development · Advanced Imaging Technologies (OCT, MRI) · Machine Learning for Biomedical Applications · Advanced Drug Disposition & Kinetics · Engineering Project Management · Medical Device Development & regulation · Project Business Case

Bachelor of Engineering – Chemical Engineering — Aug 2020 – May 2024

Thammasat University (SIIT) · Thailand · International Programme · GPA 3.22 / 4.00

- Senior Project: Computational study of butylone–β-cyclodextrin inclusion complex formation using molecular docking (AutoDock) and quantum chemical modelling (Gaussian/GaussView) (see PACCON 2024 poster below)
- Core engineering foundation: Chemical Reaction Engineering · Process Design · Plant Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics · Environmental Engineering · Wastewater Treatment · Safety in Chemical Operations · Analytical and Instrumental (XRD, FTIR, UV-Vis, Raman, XRF, NMR, AAS, ICP-OES, SEM, TEM, TGA, BET, GC, GC-MS, HPLC, LC-MS)

■ RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern – Nanomaterial Characterisation (SWCNTs) Jun 2023 – Jul 2023

Academia Sinica — Institute of Atomic and Molecular Sciences (IAMS-IIP) with Prof. Ching-Wei Lin · Taipei, Taiwan **Link**

- Selected for a competitive internship at a leading Asian materials science institute; characterised SWCNT defect populations via absorbance/fluorescence spectroscopy for cancer diagnostics
- Designed a DOE-style experimental matrix; processed multi-dimensional spectral data in Origin Lab, contributing findings to peer reports
- Achieved independent experimental operation within days of arrival, demonstrating rapid onboarding and cross-cultural adaptability

Customer Service & Operations (concurrent with full-time postgraduate studies) Aug 2024 – Jan 2026

Lily's Collection · Blue Elephant · Mi&Chi · Khao San Eatery · The Coffee Club · Auckland, NZ

- Balanced 5 concurrent part-time hospitality/retail roles while completing a full-time, research-intensive postgraduate degree, demonstrating reliability, time management, and disciplined prioritisation.
- Resolved day-to-day operational and customer issues independently across multiple fast-paced venues, requiring minimal supervision.

■ TECHNICAL SKILLS

Process Engineering: Chemical & process fundamentals; thermodynamics; heat & mass transfer; reaction engineering; fluid mechanics; process scale-up; P&ID awareness; Aspen Plus; HYSYS; HAZOP awareness

Materials & Characterisation: Microindentation; absorbance & fluorescence spectroscopy; nanomaterial characterisation; optical microscopy; mechanical property testing; thin-film material principles

Semiconductor & Mfg: Semiconductor manufacturing fundamentals; wafer fab process awareness; process control principles; yield improvement concepts; SPC; FMEA; CAPA; GMP; factory compliance

Medical Device & Regulatory: Medical device development lifecycle; Design verification & validation; ISO 13485; IEC 60601-1; IEC 62366; ISO 14971; FMEA; regulatory documentation; technical report writing; compliance; Consumer/Product Safety Assessment**;** clinical trial/study awareness

Machine Learning & Data: Python (pandas, NumPy, matplotlib, seaborn, scikit-learn); SVM classification; cross-validation; supervised ML model development & validation; Origin Lab; statistical analysis

Quality & Improvement: DOE (Design of Experiments); SPC; Six Sigma (DMAIC); Lean/Kaizen; FMEA; Fault Tree Analysis; Root Cause Analysis; Troubleshooting**;** ISO awareness

Simulation & Computational: Aspen Plus; Gaussian/GaussView; AutoDock; Discovery Studio Visualizer; molecular docking; computational chemistry; drug–material interaction modelling

Biomedical Imaging: OCT; MRI; Ultrasound — operating principles and data interpretation from dedicated postgraduate coursework

■ SOFT SKILLS

Leadership & Management: Project Management · Cross-functional Leadership · Stakeholder Management · Initiative / Self-Motivation

Problem Solving & Analysis: Critical Thinking · Analytical Thinking · Problem Solving · Troubleshooting · Attention to Detail · Continuous Improvement Mindset

Communication & Collaboration: Cross-Cultural Communication · Technical Communication · Active Listening · Conflict Resolution · Teamwork · Collaborative Mindset

Adaptability & Resilience: Adaptability · Time Management · Resilience under Pressure

■ CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Data Analysis with SQL, Excel & Power BI Specialisation (Coursera)
- KAIST Introduction to Semiconductor Process 1& 2 (Coursera)
- Google Data Analysis with Python Specialisation (Coursera)
- Poster Presentation – PACCON 2024 (Pure and Applied Chemistry International Conference)
- Academia Sinica IAMS-IIP Internship Certificate, Taiwan — international research programme

Languages: Thai (Native) · English (IELTS 6.5) · Korean (Elementary — completed KEC NZ Introductory 1A; currently King Sejong Institute Level 1B; TOPIK I target December 2026) · Japanese (Elementary)